



REGENERATIVE AGRICULTURAL INTELLIGENCE

Agro Nova

An AI assistant that walks a smallholder farmer from plot to harvest — soil, weather, diagnosis, crop choice and government support, in their own language.

India → Brazil · Russia · China

Built for national rollout, four countries

25 languages

22 Indian languages + Portuguese, Russian, Chinese

Android · iOS · Web

One Flutter codebase, every device

Smallholder farmers carry expert-level risk with none of the expert-level tools

Fragmented advice

Soil labs, weather bulletins, extension officers and input dealers each see one slice of the farm — nobody sees the whole plot.

Diagnosis by guesswork

A yellowing leaf can be nutrient deficiency, pest or disease. Wrong guesses mean wasted chemicals and lost yield.

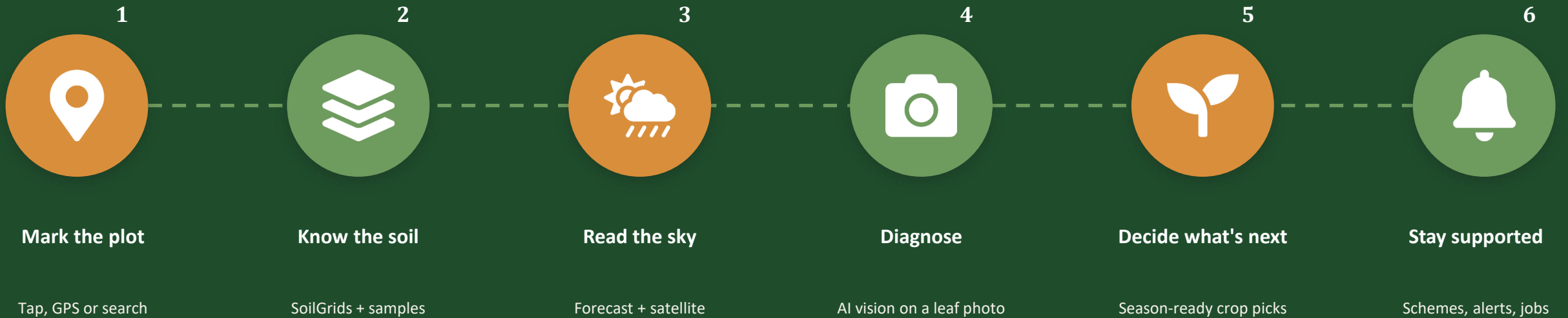
Support that doesn't reach them

Government schemes, subsidies and weather alerts exist, but rarely arrive in the right language at the right moment.

Agro Nova puts a single, plot-aware assistant in the farmer's pocket — grounded in real soil, satellite and weather data, and available even offline.

HOW IT WORKS

One journey, from first tap to harvest





FEATURE 01

Plot Capture

Shapely + pyproj

Server-side shape, area and country validation

Every recommendation starts with a precise, validated field boundary.

- **Four ways to mark a plot**

Tap the map, drop a GPS pin, type coordinates, or search a place — whichever suits the farmer and the signal.

- **Validated, not just drawn**

The backend checks shape, area and country against the polygon before it becomes the plot of record.

- **The anchor for everything else**

Soil, weather, satellite and crop advice are all pulled for this exact geometry, not a generic district average.



FEATURE 02

Soil Intelligence

ISRIC SoilGrids

Public global soil data, free at any plot

Public data first, ground truth whenever it's available.

- **Instant public baseline**

ISRIC SoilGrids gives every plot a soil profile the moment it's mapped — no lab visit required to get started.

- **Room for real samples**

Farmers can enter government or lab test values against a GPS point, sharpening the picture over time.

- **A path to better data**

The app explains how and where to get soil tested locally, closing the loop between public and ground data.



FEATURE 03

Weather & Climate Outlook

3 satellites

CHIRPS · ERA5-Land · SMAP via Google Earth Engine

A crop-specific outlook, not a generic weather app.

- **Forecast meets satellite**

A 10-day forecast is layered with NDVI, rainfall-against-normal and soil moisture from Earth Engine.

- **Falls back gracefully**

Open-Meteo steps in automatically if Earth Engine or another source is unavailable — the farmer always sees an outlook.

- **El Niño / La Niña aware**

ENSO effects are factored in per country, so the outlook reflects the season the farmer is actually in.



FEATURE 04

AI Plant Diagnosis

3-way AI fallback

OpenAI → Anthropic → Gemini, whichever
answers first

A leaf photo becomes a diagnosis and an organic remedy in one step.

- **Vision-model triage**

A photo of a leaf is read by whichever configured AI vision model is available and answers first.

- **Deficiency, pest or disease**

The model identifies the likely cause — the hardest call for a farmer to make from symptoms alone.

- **Organic remedy, explained**

Preparation steps come with a plain explanation of why organic outperforms chemical for that case.



FEATURE 05

What To Grow Next

Kharif · Rabi · Zaid

Recommendations tied to the next local sowing
season

Crop choice grounded in the plot, not a generic almanac.

- **Four signals, one recommendation**

AI recommendations combine the plot's soil, satellite data, ENSO state and location.

- **Season-aware**

Suggestions target the next local sowing window — Kharif, Rabi, Zaid and equivalents in other countries.

- **Built to travel**

The same reasoning extends beyond India's seasons as the app reaches Brazil, Russia and China.

Resilience, Water & Value Addition

A generic guide for every farm, plus an AI section personalised to this plot, crop and livestock.



Compost

Build and manage on-farm compost to restore soil organic matter.



Biogas

Turn farm and livestock waste into household cooking fuel.



Milk & the cow calculator

Dairy guidance plus a calculator sized to the farmer's herd.



Water saving

Practical tips to cut irrigation waste, personalised to the crop.



FEATURE 07

Schemes & Smart Alerts

FCM push

Scheme, weather and market alerts, tap-to-open

Government support and timely nudges, delivered, not just published.

- **A living schemes catalogue**

Government schemes and subsidies are catalogued and kept in reach inside the app.

- **Three kinds of push**

Scheme, weather and market notifications go out through Firebase Cloud Messaging.

- **Jobs that finish quietly**

Slow AI answers run as background jobs; a farmer who leaves gets a “ready” notification that opens the finished answer.

Built for scale, from day one



25
languages

22 Indian languages plus Portuguese, Russian and Chinese — screen labels are built in; for widely used languages the AI writes advice directly in the farmer's language, and the Cloud Translation API covers and caches the rest.



4
countries

Launching for smallholder farmers in India, and built from the ground up to extend to Brazil, Russia and China — the same Flutter code, on Android, iOS and the web.

A backend built to stay up and stay honest



API & data

Python 3.12, FastAPI, Pydantic, SQLAlchemy. SQLite locally, Postgres in production, Alembic migrations. Tested with pytest across SQLite, Postgres and Redis.



AI, resilient by design

Gemini , OpenAI and Anthropic called in a configurable order, each behind retries and a circuit breaker — one answer always gets through.



Satellite & scale-out

earthengine-api and firebase-admin power satellite reads; optional Redis shares caches and rate limits; Cloud Tasks runs jobs on any instance; a nightly job precomputes satellite data per plot.



Observability & deploy

Prometheus metrics, alert rules and a local Grafana dashboard ship with the service, designed to run on Cloud Run.

Private, compliant, and useful without signal



Privacy, DPDP-ready

- Versioned privacy notice shown up front
- Separate consent for plots & farm details, AI advice, and notifications — as India's DPDP Act requires
- Data download and erasure on request
- Inactive accounts erased after a set retention period



Offline-first

- Saved answers stay visible with no signal at all
- Plots and soil samples queue locally and sync later
- Sync is de-duplicated — no repeated submissions
- Push-when-ready notifications open the finished answer the moment signal returns

One Flutter codebase, every device, fails soft



Flutter + Provider, `google_maps_flutter`, geolocator and `image_picker` power the map, GPS and camera flows across Android, iOS and the web from a single codebase.



Firebase Auth, Messaging and App Check, alongside a no-credentials dev login — every Google service is optional and fails soft if unavailable.



An Earth Engine to BigQuery pipeline computes climatology today, laying the groundwork for a future Vertex AI seasonal model.



Grown from the plot up.

Agro Nova — regenerative agricultural intelligence for smallholder farmers, from India to the world.